

Diagnostic Accuracy of Fluorescence Immunoassay (FIA) versus Enzyme-Linked Immunosorbent Assay (ELISA) for Dengue Virus NS1 Antigen Detection: A Systematic Review and Meta-Analysis

Running title: FIA vs ELISA for dengue NS1: a DTA meta-analysis

Article type: Systematic review and meta-analysis of diagnostic test accuracy

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PROSPERO: CRD[XXXXXXX] (pending assignment) · Protocol version 1.0, August 2026

Word count: Abstract ~280 · Main text (target after data lock: 4,500–6,000)

Figures: 8 · **Tables:** 13 · **Supplementary files:** S1–S6

Keywords: dengue; NS1 antigen; fluorescence immunoassay; FIA; ELISA; diagnostic accuracy; sensitivity; specificity; meta-analysis; QUADAS-2; HSROC; bivariate model; point-of-care testing

Abstract

Background. Dengue virus (DENV) infection is the most rapidly expanding mosquito-borne viral disease globally, with more than 14.4 million cases reported to the World Health Organization in 2024. Early detection of non-structural protein 1 (NS1) antigen during the acute febrile phase (days 1–7) is critical for timely care. Enzyme-linked immunosorbent assay (ELISA) is accurate but requires laboratory infrastructure and 4–8 hours per batch. Fluorescence immunoassay (FIA) provides reader-based, quantitative or semi-quantitative results within 15 minutes; its accuracy relative to ELISA has not been synthesized as a distinct technology class.

Objectives. To estimate pooled sensitivity, specificity, positive and negative likelihood ratios (PLR, NLR), and diagnostic odds ratio (DOR) of FIA for dengue NS1; to compare FIA with ELISA on paired specimens; and to explore heterogeneity by infection status, fever duration, serotype, and platform.

Methods. PubMed/MEDLINE, Embase, Web of Science, Scopus, ScienceDirect, Cochrane Library, and Google Scholar were searched from inception to September 2026 without language restrictions. Eligible studies enrolled patients with suspected acute dengue (fever ≤ 7 days) and reported or permitted reconstruction of 2×2 tables. Risk of bias was assessed with a dengue-tailored QUADAS-2 tool. Pooled estimates used the bivariate random-effects model of Reitsma (equivalent, without covariates, to the hierarchical summary ROC [HSROC])

model of Rutter and Gatsonis). Heterogeneity was quantified with Cochran's Q , I^2 , and bivariate τ^2 . Publication bias was assessed with Deeks' funnel plot asymmetry test. Reporting followed PRISMA-DTA.

Results. *[To be completed after dual screening, extraction, and bivariate modelling. Report k , N , pooled sensitivity and specificity with 95% CIs, PLR, NLR, DOR, HSROC AUC, heterogeneity, paired FIA vs ELISA contrast, and key subgroups.]*

Conclusions. *[To be completed after analysis. State whether FIA is appropriate as a near-point-of-care rule-in test, whether a negative FIA can exclude dengue, and the residual role of ELISA/RT-PCR.]*

Registration. PROSPERO CRD[XXXXXXX].

1. Introduction

1.1 Global burden of dengue

Dengue virus (DENV) infection is among the most important arthropod-borne viral threats of the twenty-first century. Four antigenically distinct serotypes (DENV-1–4) of a single-stranded positive-sense RNA flavivirus are transmitted mainly by *Aedes aegypti* and *Aedes albopictus* [1,2]. Bhatt et al. estimated 390 million infections per year (95% credible interval 284–528 million), of which 96 million (67–136 million) are apparent at any severity—more than three times contemporaneous WHO figures [3]. Asia bore 70% of that burden; India alone accounted for about one-third of apparent infections [3].

In 2024 the recorded burden reached a historic high: 14,434,584 cases, including 7,718,585 laboratory-confirmed infections, 52,738 severe cases, and 11,201 deaths, across all six WHO regions and more than 100 countries [4]. The Region of the Americas contributed over 90% of reports; Brazil alone exceeded 10 million cases and 6,321 deaths [4]. Climate-driven expansion of *Aedes* habitats, urbanization, mobility, and shifting serotype immunity underlie this surge [5,6]. WHO lists dengue among its leading global health threats [5].

1.2 NS1 as an acute-phase biomarker

Clinical dengue—undifferentiated fever, myalgia, retro-orbital pain, rash—overlaps malaria, chikungunya, Zika, leptospirosis, and typhoid [7,8]. Syndromic diagnosis is therefore unreliable; laboratory confirmation is required for care, outbreak detection, and surveillance [2].

NS1 is a 46–50 kDa conserved glycoprotein secreted at up to ~50 $\mu\text{g/mL}$ during peak viremia [9,10]. Antigenemia is detectable from day 1 of symptoms, 2–4 days before IgM, and often through day 9 [9,11]. In primary infection, detection exceeds 80% in the first four days, with peak levels on days 1–4 [12]. Serum NS1 correlates with RT-PCR viremia [9,13].

In secondary infection, anamnestic anti-NS1 IgG sequesters free NS1 into immune complexes, shortening detectability and reducing sensitivity to about 50–63% in the first four days [12,14]. Serotype also matters: primary DENV-1 typically yields higher NS1 than primary DENV-2 (early detection sometimes 50–67%) [12]. Platforms must therefore perform across immune status, serotype, and day of illness.

1.3 ELISA: laboratory standard and operational limits

NS1 ELISA remains the recommended laboratory method where infrastructure exists [2,15]. Platelia Dengue NS1 Ag-ELISA (Bio-Rad) and Panbio Dengue Early ELISA (Abbott) use sandwich formats with peroxidase-labelled monoclonal antibodies [16,17].

Pooled specificity is high (97–99%); sensitivity varies. Da Silva et al. (30 studies, 12,105 patients) reported Platelia sensitivity 74% (63–82%) and Panbio 66% (61–71%), falling to 66% and 57% in secondary infection [18]. Guzmán et al., in a six-country evaluation, found Platelia more sensitive than Pan-E (66% vs 52%) with 100% specificity in non-dengue febrile patients [19]. Pillay et al. (161 studies; Bayesian random-effects without assuming a perfect gold standard) estimated NS1 ELISA sensitivity 90% (95% CrI 68–98%) on days 0–4, approaching RT-PCR (95%; 77–99%) [20].

ELISA nonetheless requires plate readers, incubators, trained staff, 2–8 °C storage, and batching, with 4–8 hours from sample receipt [15,21,22]. Transport from primary or rural sites may add 6–24 hours, so time-to-result often exceeds 24–48 hours [22]. The critical phase of severe dengue clusters around days 3–7, when NS1 is already declining [2,7]. Batch queues also stall individual results in low-volume periods and overwhelm laboratories during surges [21].

1.4 Fluorescence immunoassay

FIA aims to combine ELISA-like objective readout with point-of-care speed. Unlike visual immunochromatographic tests (ICTs), FIA uses fluorescent labels and a compact optical reader to report a cutoff index (COI) [23,24]. STANDARD F Dengue NS1 Ag FIA (SD Biosensor) uses a barcoded cartridge on the F200 analyzer and returns results in 5–15 minutes from 100 µL of whole blood, serum, or plasma [23,25]. The NS1 Ag QUANTI CARD uses a related fluorescence principle [26].

Pohekar and Bhalchandra (n = 300, India) found STANDARD F essentially equivalent to NS1 ELISA, with an inverse COI–platelet relationship that is hypothesis-generating for severity [27]. Ghogre reported QUANTI CARD sensitivity 96.5% and specificity 96.4% versus ELISA, versus ICT 65.5%/83.9%, with near-perfect FIA–ELISA agreement [26]. Chaiyo et al., against combinatorial RT-PCR plus IgM/IgG ELISA, reported STANDARD F NS1 sensitivity 79.1% and specificity 92.3%, slightly above paired ICT (76.6%/92.3%) [25].

1.5 Evidence gap

Published FIA sensitivity ranges from about 79% to 97% and specificity from 92% to 96%, reflecting mix of primary/secondary infection, serotype, sampling day, reference standard, geography, and brand [25–27]. Prior syntheses did not isolate FIA. Mata et al. (PROSPERO CRD42014009885; 57 studies, 40 in meta-analysis) reported NS1/IgM/IgG ICT sensitivity 91% and specificity 96% without separating FIA from visual ICT [28]. Haider et al. (15 studies, 4,135 participants) found NS1-only ICT sensitivity 70.97% and specificity 94.73% (DOR 43.95) [29]. Frazer and Norton reviewed dengue diagnostics in the vaccine era but did not pool FIA [30].

FIA and visual ICT differ in signal (fluorescence vs colloidal gold), interpretation (COI vs lines), analytical limit, and operator dependence. Pooling them obscures FIA's true performance and misleads procurement.

1.6 Hypotheses and objectives

Primary hypothesis. FIA for dengue NS1 is diagnostically comparable to ELISA, with a turnaround time suitable for near-point-of-care use.

Secondary hypotheses. (i) Sensitivity is higher in primary than secondary infection. (ii) Sensitivity is higher on days 0–3 than days 4–7. (iii) Accuracy varies by serotype (higher for DENV-1/3 than DENV-2/4). (iv) Reference-standard choice (RT-PCR vs ELISA vs composite) changes pooled estimates.

Primary objective. To estimate pooled sensitivity, specificity, PLR, NLR, and DOR of FIA for dengue NS1 in suspected acute dengue, using a bivariate random-effects meta-analysis.

Secondary objectives. (1) Head-to-head FIA vs ELISA on the same specimens. (2) HSROC curve and AUC. (3) Heterogeneity and pre-specified subgroups/meta-regression (infection status, fever day, serotype, reference, platform, region). (4) QUADAS-2 and quality-restricted sensitivity analysis. (5) Deeks’ test. (6) Algorithm recommendations for resource-limited and point-of-care settings.

2. Methods

This diagnostic test accuracy (DTA) review was designed and reported according to PRISMA-DTA [31], the PRISMA-DTA explanation and elaboration [32], PRISMA 2020 [33], PRISMA-S [34], and Cochrane DTA guidance [35]. STARD 2015 informed assessment of primary-study reporting [36]. A PRISMA-DTA checklist is Supplementary Table S1. Ethical approval was not required (secondary analysis of published data).

2.1 Protocol and registration

The protocol was registered with PROSPERO before formal screening and extraction (CRD[XXXXXXX], pending; version 1.0, August 2026). Deviations will be dated, justified, and reported.

2.2 Review question (PIRD)

What is the diagnostic accuracy of fluorescence immunoassay (FIA) compared with enzyme-linked immunosorbent assay (ELISA) for detection of dengue virus NS1 antigen in patients with suspected acute dengue infection?

Component	Definition
Population	Any age; clinically suspected acute dengue; fever ≤ 7 days from onset; any healthcare setting in endemic or epidemic regions. Specimens: venous whole blood, serum, or plasma.
Index test	Commercial FIA for dengue NS1 (e.g., STANDARD F, QUANTI CARD, or equivalent reader-interpreted fluorescence assays). Positivity: manufacturer COI.
Reference standard	Tier 1: RT-PCR; Tier 2: virus isolation $\pm$ RT-PCR; Tier 3: composite (RT-PCR + paired IgM/IgG seroconversion $\pm$ NS1 ELISA); Tier 4: NS1 ELISA alone if molecular testing is unavailable. ELISA is also the comparator in paired analyses.
Target condition	Acute DENV-1–4 infection confirmed by NS1 antigenemia and/or virological evidence in the febrile window.

2.3 Eligibility criteria

Inclusion. Patients of any age with suspected acute dengue and fever ≤ 7 days (explicit criterion or documented mean/median ≤ 7 days). FIA for NS1 with automated or semi-automated COI. Reference: RT-PCR, isolation, composite, and/or NS1 ELISA. Designs: cross-sectional DTA, prospective or retrospective diagnostic cohorts, head-to-head comparisons, nested surveillance. Extractable or reconstructable 2×2 table (true positives [TP],

false positives [FP], false negatives [FN], true negatives [TN]). Original data; peer-reviewed articles, data-sufficient abstracts, and preprints (preprints flagged for sensitivity analysis). No language or date limits.

Exclusion. Asymptomatic seroprevalence or blood-bank screening only; convalescent-only specimens (>7 days) without paired acute samples; animal or spiked-sample studies without a clinical cohort; saliva/urine only unless paired with a standard matrix; combined NS1/IgM/IgG assays that cannot be disaggregated; exclusively immunocompromised cohorts without a general arm; archived specimens without clinical context; reviews, editorials, case reports/series without denominators; non-reconstructable 2×2 data; duplicate datasets (most complete report retained).

2.4 Information sources and search

Searches ran from inception to September 2026 (updated immediately before final analysis) in PubMed/MEDLINE, Embase, Web of Science Core Collection (SCI-EXPANDED, ESCI), Scopus, ScienceDirect (title–abstract–keyword), Cochrane Library (CDSR, CENTRAL), and Google Scholar (first 200 by relevance). Grey sources: ClinicalTrials.gov, WHO ICTRP, ProQuest Dissertations, medRxiv/bioRxiv, WHO IRIS, PAHO guidance, manufacturer IFUs, and backward/forward citation tracking.

The strategy combined five blocks: dengue AND NS1 AND (FIA OR ELISA) AND diagnostic-accuracy terms. Full database strings are Supplementary File S2. Index (PubMed) strategy:

```
("Dengue"[MeSH] OR "Dengue Virus"[MeSH] OR "Severe Dengue"[MeSH]
OR dengue[tiab] OR DENV[tiab] OR "dengue fever"[tiab]
OR "dengue hemorrhagic fever"[tiab] OR "dengue shock syndrome"[tiab])
AND
("Viral Nonstructural Proteins"[MeSH] OR NS1[tiab] OR "NS-1"[tiab]
OR "NS1 antigen"[tiab] OR "nonstructural protein 1"[tiab]
OR "non-structural protein 1"[tiab])
AND
(("Fluoroimmunoassay"[MeSH] OR "fluorescence immunoassay"[tiab]
OR "fluorescent immunoassay"[tiab] OR FIA[tiab] OR TRFIA[tiab]
OR "STANDARD F"[tiab] OR "SD Biosensor"[tiab] OR "QUANTI CARD"[tiab])
OR
("Enzyme-Linked Immunosorbent Assay"[MeSH] OR ELISA[tiab]
OR "enzyme immunoassay"[tiab] OR Platelia[tiab] OR Panbio[tiab]))
AND
("Sensitivity and Specificity"[MeSH] OR "Predictive Value of Tests"[MeSH]
OR "ROC Curve"[MeSH] OR sensitivity[tiab] OR specificity[tiab]
OR "diagnostic accuracy"[tiab] OR "likelihood ratio"[tiab]
OR "true positive"[tiab] OR "false negative"[tiab])
```

The search was validated by retrieval of known eligible reports [20,25–27] and developed with PRESS 2015 principles [37].

2.5 Study selection

Records were exported to Zotero or EndNote, deduplicated, and screened in Rayyan or Covidence. Two reviewers independently screened titles/abstracts and full texts after a 50-record calibration (proceed when Cohen's $\kappa \geq 0.80$) [38]. Unclear abstracts advanced to full text. Full-text exclusions used a taxonomy: wrong population; wrong index test; no 2×2 data; wrong target condition; duplicate; other. Disagreements were resolved by discus-

sion, then a third reviewer. Reviewers were not blinded to journal or authors. Counts followed a PRISMA 2020 flow diagram.

2.6 Data collection

A piloted form (Supplementary File S3) was applied independently by two reviewers to every included study. Authors were emailed once plus a 14-day reminder for missing 2×2 data. Missing items were coded NR and not imputed except the pre-specified continuity correction.

Extracted items: study identifiers, design, setting, funding; N, prevalence, age, sex, fever day, WHO 2009 class [2], primary/secondary status and method, serotypes, specimen; FIA brand, reader, COI, IFU cutoff, volume, turnaround, blinding, indeterminates; reference type, ELISA/RT-PCR details, same-specimen testing, interval, differential verification; 2×2 cells and derived accuracy.

When cells were unreported: $TP = Se \times n_{\text{positive}}$; $FN = (1 - Se) \times n_{\text{positive}}$; $TN = Sp \times n_{\text{negative}}$; $FP = (1 - Sp) \times n_{\text{negative}}$ (rounded to integers). Non-reconstructable studies were described narratively only.

2.7 Risk of bias (QUADAS-2)

QUADAS-2 [39] was tailored to dengue NS1 FIA (Supplementary File S4). Two reviewers assessed studies after a three-study calibration.

Table 1. Tailored QUADAS-2 signalling questions

Domain	Questions (standard and review-specific)
1. Patient selection	Consecutive or random sample? Case-control avoided? Inappropriate exclusions avoided? <i>Fever ≤ 7 days at draw?</i>
2. Index test (FIA)	Interpreted without knowledge of the reference? Threshold pre-specified? <i>Performed per IFU? Indeterminates reported and handled (<5% excluded)?</i>
3. Reference standard	Likely to classify acute dengue correctly? Interpreted without knowledge of FIA? <i>Incorporation bias avoided? Applied in its diagnostic window?</i>
4. Flow and timing	Interval appropriate (same specimen or ≤ 24 h)? Same reference for all? All patients analysed? <i>Flow described (STARD-style)?</i>

Judgments: Low / High / Unclear for risk of bias; Domains 1–3 also for applicability. High RoB examples: two-gate sampling; post-hoc COI optimization; IgM-only reference; differential verification; >5% unexplained exclusions. NS1 ELISA as sole reference was Unclear (imperfect standard), not automatically High. Visualization used `robvis(tool = "QUADAS-2")` [40]. Overall RoB entered sensitivity analysis and meta-regression.

2.8 Statistical synthesis

Outcomes. Pooled sensitivity, specificity, $PLR = Se/(1 - Sp)$, $NLR = (1 - Se)/Sp$, DOR, and HSROC (AUC, 95% confidence and prediction regions). Secondary: paired FIA vs ELISA, Cohen’s κ , narrative turnaround time.

Primary model. Bivariate random-effects GLMM (Reitsma) in `Rmada::reitsma()` [41,42], jointly modelling logit sensitivity and logit false-positive rate with unstructured Σ (τ^2_{Se} , τ^2_{FPR} , correlation ρ). Variance: REML (ML in sensitivity analysis). PLR, NLR, and DOR 95% CIs: parametric bootstrap (10,000 draws). Without covariates this model is equivalent to HSROC (Rutter and Gatsonis) [43,44]. Zero cells: +0.5 to all four cells [35,41]; 0.1 and 1.0 in sensitivity analysis.

Head-to-head. Separate bivariate models on the overlapping study set when both tests were applied to the same specimens; ≥ 4 studies required.

Publication bias. Deeks' regression of $\ln(\text{DOR})$ on $1/\sqrt{\text{ESS}}$, where $\text{ESS} = 4 \times n_{\text{pos}} \times n_{\text{neg}} / (n_{\text{pos}} + n_{\text{neg}})$; two-sided $p < 0.10$ indicated possible small-study effects [45].

Software. $R \geq 4.3.0$ [46] with `mada` [41], `metafor` [47], `meta`, `mvmeta`, `robvis` [40], `ggplot2`. Script: Supplementary File S5.

2.9 Heterogeneity and subgroups

Heterogeneity: forest/HSROC inspection; Cochran's Q ($\alpha = 0.10$) [48]; I^2 (0–40% might not be important; 30–60% moderate; 50–90% substantial; 75–100% considerable) [48]; bivariate τ^2 ; $\rho < -0.3$ suggesting a threshold effect.

Subgroup bivariate models required ≥ 4 studies per stratum: primary vs secondary [14]; days 0–3 vs 4–7 [9]; DENV-1–4 [12]; reference tier; STANDARD F vs other FIA; region; pediatric vs adult; prospective vs retrospective. Mixed (unstratified) data stayed in the overall model only.

2.10 Sensitivity analyses and meta-regression

Pre-specified: (1) exclude High RoB in ≥ 2 domains; (2) RT-PCR/isolation reference only; (3) leave-one-out; (4) peer-reviewed full texts only; (5) continuity corrections 0.1 and 1.0; (6) ML vs REML.

Univariable bivariate meta-regression (`cbind(tsens, tfpr) ~ covariate`) for reference type, brand, specimen, design, prevalence ($\leq 50\%$ vs $> 50\%$), region, fever category, infection mix, and overall RoB. Nested ML models compared by likelihood-ratio test ($p < 0.05$). GRADE for DTA [49] was applied if the evidence base allowed (Supplementary Table S6).

2.11 Patient involvement and data sharing

Patients were not involved in protocol design. Upon acceptance, 2×2 data, QUADAS-2 judgments, R code, and the PRISMA-DTA checklist will be deposited in a public repository.

3. Results

Production note. Quantitative cells below are locked to dual-reviewer extraction and `mada` output. They must not be invented. After analysis, replace bracketed tokens and insert Figures 1–8.

3.1 Study selection

The search was run on [DD Month 2026] (update: [date or NA]). Hits by source are in Table 2.

Table 2. Records by information source

Source	Records (n)
PubMed/MEDLINE	[n]
Embase	[n]

Source	Records (n)
Web of Science Core Collection	[n]
Scopus	[n]
ScienceDirect	[n]
Cochrane Library	[n]
Google Scholar (first 200)	[n]
Trial registries, theses, preprints, citations, IFUs	[n]
Total before deduplication	[N_total]
Duplicates removed	[n_dup]
Unique records screened	[N_unique]

Title/abstract exclusions: [n_ta_excl]. Full texts assessed: [n_ft]; not retrieved: [n]. Full-text exclusions [n_ft_excl]: wrong population [n]; wrong index test [n]; no 2×2 [n]; wrong target condition [n]; duplicate [n]; other [n].

Included in qualitative synthesis: [k_qual]; in FIA bivariate meta-analysis: [k_fia]; paired FIA–ELISA: [k_h2h]. Inter-rater κ: title/abstract [x.xx]; full text [x.xx]. Author contact: [n] studies; [n] supplied additional cells.

Figure 1. PRISMA 2020 flow diagram.

3.2 Study characteristics

[k_qual] studies ([year_min]–[year_max]) contributed [N_patients] participants ([n+] dengue-positive, [n–] negative; prevalence [xx.x%]). Designs: [n] prospective, [n] retrospective, [n] other. Multicenter: [n].

Table 3. Characteristics of included studies

Study	Country	Design & setting	N	Fever days	Primary/secondary	Serotypes	FIA	ELISA	Reference	Specimen
[Author, year]										
Total	[k] countries		[N]							

Geography (Figure 2). South Asia [k]; Southeast Asia [k]; Western Pacific [k]; Latin America [k]; Africa [k]; Eastern Mediterranean [k]; other [k].

Demographics. Age mix [n]; pediatric-only [n]; adult-only [n]; age range [x]–[y] years. Male [x%]–[y%]. Fever-day restriction 0–3: [n]; 0–7: [n]; day-stratified 2×2: [n]. Primary [n_prim]; secondary [n_sec]. Serotypes: DENV-1 [n], DENV-2 [n], DENV-3 [n], DENV-4 [n]. Specimens: serum [k], plasma [k], whole blood [k].

Index and reference. STANDARD F [k]; QUANTI CARD [k]; other FIA [k]. Manufacturer COI [k]; modified [k]. ELISA: Platelia [k]; Panbio [k]; STANDARD E [k]; other [k]. Reference tiers: RT-PCR [k]; isolation [k]; composite [k]; ELISA alone [k]. Same-specimen testing [k].

Table 4. FIA 2×2 tables versus reference standard

Study	TP	FP	FN	TN	N	Se (%)	Sp (%)	Zero-cell
[Author, year]								Y/N
Pooled (bivariate)	—	—	—	—	[N]	§3.4	§3.4	—

ELISA 2×2 and FIA–ELISA agreement tables: Supplementary Tables S4–S5. Scoping candidates (Pohekar 2023, Ghogre 2021, Chaiyo 2022) enter only if they pass dual screening.

3.3 Risk of bias

QUADAS-2 κ (overall) = [x.xx]. Traffic-light and summary plots: Figures 3–4. Full judgements: Supplementary Table S6.

Table 5. QUADAS-2 summary (k = [k])

Domain	Low n (%)	High n (%)	Unclear n (%)
D1 Patient selection	[n] ([%])	[n] ([%])	[n] ([%])
D2 Index test	[n] ([%])	[n] ([%])	[n] ([%])
D3 Reference standard	[n] ([%])	[n] ([%])	[n] ([%])
D4 Flow and timing	[n] ([%])	[n] ([%])	[n] ([%])
Overall	[n] ([%])	[n] ([%])	[n] ([%])

Domain synthesis. [Patient selection: consecutive vs convenience, fever-day reporting, spectrum. Index test: blinding/automated COI, pre-specified cutoff, IFU, indeterminates. Reference: RT-PCR vs ELISA-only, incorporation bias, diagnostic window. Flow: same specimen, differential verification, exclusions.] Overall Low RoB [n]; High in ≥ 2 domains [n]; Unclear [n].

Figure 3. QUADAS-2 traffic-light plot.

Figure 4. QUADAS-2 weighted summary bars.

3.4 Diagnostic accuracy

3.4.1 FIA versus reference (primary model)

Bivariate REML model: [k_fia] studies, [N] participants; [n] studies with 0.5 continuity correction.

Table 6. Pooled FIA accuracy for dengue NS1 (bivariate GLMM)

Measure	Estimate	95% CI
Sensitivity	[xx.x%]	[ll.l]–[uu.u]%
Specificity	[xx.x%]	[ll.l]–[uu.u]%
PLR	[x.xx]	[ll]–[uu]
NLR	[0.xxx]	[ll]–[uu]
DOR	[xxx]	[ll]–[uu]
HSROC AUC	[0.xxx]	[ll]–[uu]

Heterogeneity: Q_{Se} [Q] (df [k-1], $p = [p]$); I^2_{Se} [xx.x%]; τ^2_{Se} [x.xxxx]. Q_{Sp} [Q]; I^2_{Sp} [xx.x%]; τ^2_{FPR} [x.xxxx]. $\rho = [-0.xx]$ (threshold effect: [yes if $\rho < -0.3$ / no / weak]). HSROC: Λ [x.xx], Θ [x.xx], β [x.xx].

Figure 5. Coupled forest plots (FIA sensitivity and specificity).

Figure 6. HSROC curve with 95% confidence and prediction regions.

3.4.2 ELISA versus independent reference

If $k \geq 4$: pooled ELISA Se [xx.x%] ([ll]–[uu]), Sp [xx.x%] ([ll]–[uu]), PLR [x.xx], NLR [0.xxx], DOR [xxx], AUC [0.xxx] ($k = [k_{elisa}]$, $N = [N]$). If $k < 4$: narrative range only.

3.4.3 Paired FIA versus ELISA

[k_h2h] studies on the same specimens.

Table 7. Head-to-head bivariate estimates (identical study set)

Test	k	N	Sensitivity (95% CI)	Specificity (95% CI)	PLR	NLR
FIA (subset)	[k]	[N]	[xx.x%] ([ll]–[uu])	[xx.x%] ([ll]–[uu])	[x.xx]	[0.xxx]
ELISA	[k]	[N]	[xx.x%] ([ll]–[uu])	[xx.x%] ([ll]–[uu])	[x.xx]	[0.xxx]
Difference (FIA – ELISA)	—	—	[Δ Se, pp]	[Δ Sp, pp]	—	—

Agreement: [xx.x%]; κ [0.xx] ([ll]–[uu]); McNemar $p = [p]$.

Figure 7. Comparative HSROC (FIA vs ELISA).

3.5 Subgroups and meta-regression

Models required ≥ 4 studies per stratum.

Table 8. FIA by fever day and infection status

Stratum	k	N	Se (95% CI)	Sp (95% CI)	PLR	NLR	DOR
Days 0–3	[k]	[n]					
Days 4–7	[k]	[n]					
Primary	[k]	[n]					
Secondary	[k]	[n]					

LR tests: fever category $\chi^2 = [x.xx]$, df [d], $p = [p]$; infection status $\chi^2 = [x.xx]$, $p = [p]$.

Table 9. Additional subgroups (FIA)

Covariate	Stratum	k	Se (95% CI)	Sp (95% CI)	Fitted (≥ 4)?
Serotype	DENV-1 / 2 / 3 / 4	[k]			Y/N
Reference	RT-PCR / ELISA / composite	[k]			Y/N
Platform	STANDARD F / other	[k]			Y/N
Region	S Asia / SE Asia / LatAm / other	[k]			Y/N

Covariate	Stratum	k	Se (95% CI)	Sp (95% CI)	Fitted (≥4)?
Age	≤18 y / >18 y	[k]			Y/N
Design	Prospective / retrospective	[k]			Y/N

Table 10. Univariable bivariate meta-regression (ML likelihood-ratio tests)

3.7 Principal numerical findings

In [k] studies ([N] participants) from [regions], FIA for dengue NS1 showed pooled sensitivity [xx.x%] (95% CI [ll]–[uu]) and specificity [xx.x%] ([ll]–[uu]), with PLR [x.xx], NLR [0.xxx], DOR [xxx], and HSROC AUC [0.xx]. In [k] paired studies, FIA [was similar to / differed from] ELISA by [Δ Se] pp and [Δ Sp] pp. Heterogeneity was [I^2 interpretation]. Accuracy [was / was not] lower in secondary infection and [was / was not] higher in the first 3 days of fever. Deeks' test [did / did not] indicate small-study effects ($p = [p]$). Sensitivity analyses [supported / qualified] the primary operating point.

4. Discussion

4.1 Principal findings in context

This review is, to our knowledge, the first DTA synthesis to isolate FIA for dengue NS1 rather than pooling it with visual ICT. Across [k] studies and [N] participants, FIA pooled sensitivity was [xx.x%] (95% CI [ll]–[uu]) and specificity [xx.x%] ([ll]–[uu]), with PLR [x.xx], NLR [0.xxx], DOR [xxx], and AUC [0.xx]. On paired specimens ([k_h2h] studies), FIA differed from ELISA by [Δ Se] pp (sensitivity) and [Δ Sp] pp (specificity).

Visual NS1 ICT pooled sensitivity is about 71% with specificity near 95% (Haider et al.: 70.97%/94.73%, DOR 43.95) [29]; combination NS1/IgM/IgG ICT approached 91%/96% with $I^2 > 90\%$ [28]. Single-centre FIA studies suggested reader-based fluorescence can exceed visual ICT on the same samples [25,26]. *[After data: if pooled FIA Se exceeds ~71% with similar or higher Sp, FIA closes part of the ICT–ELISA gap; otherwise it remains in the ICT envelope.]*

ELISA remains the laboratory workhorse (Platelia Se 74%, Panbio 66%, both Sp ~99% [18]; Platelia 66% vs Pan-E 52% [19]; Bayesian NS1 ELISA Se 90% on days 0–4 [20]; WHO/TDR NS1 ELISA Se 60–75% vs RT-PCR, worse in secondary infection [17]). *[After paired results: overlapping CIs mean interchangeability is not excluded; separated CIs with higher ELISA Se mean FIA is a rapid adjunct, not a laboratory replacement.]*

A PLR [x.xx] produces a [large / moderate / small] rise in post-test probability when FIA is positive; an NLR [0.xxx] [does / does not] reliably exclude dengue [50]. In high-prevalence outbreaks, PPV rises even with a moderate PLR, while the same NLR misses more cases. That asymmetry supports FIA as a **rule-in** test in the febrile window, with negatives reflexed to ELISA, RT-PCR, or paired serology [2,7].

I^2 was [xx%] (sensitivity) and [xx%] (specificity). Subgroups [supported / did not support] lower Se in secondary infection and higher Se on days 0–3, consistent with NS1 kinetics [9,12,14]. Deeks' $p = [p]$ [45].

FIA therefore occupies a middle stratum: faster than ELISA, more objective than visual ICT, and [**comparable to / modestly inferior to / not distinguishable from**] ELISA on paired specimens, at the cost of a reader, cartridges, and residual sensitivity loss in secondary and late-acute infection.

4.2 Clinical and operational implications

WHO 2009 guidance ties risk stratification to defervescence (typically days 3–7) [2]. NS1 peaks in the first four days, often before IgM [9,30]. ELISA plus transport commonly costs 24–48 hours [15,22]—enough to span the critical phase. FIA in 5–15 minutes [23,25,27] changes *when* fluid management and admission can be informed.

In endemic emergency departments the differential includes malaria, leptospirosis, typhoid, chikungunya, and Zika [7,8]. A positive FIA (PLR [x.xx]) can start dengue-protocol monitoring without waiting for ELISA. COI–platelet correlation remains hypothesis-generating [27]. Because NS1 sensitivity falls in secondary infection and after day 3–4 [9,14,17], a negative FIA in a patient with warning signs or prior dengue **must not** stop the work-up; add IgM/IgG or RT-PCR [28,29]. Automated COI reduces visual-line disagreement [17,39] if the manufacturer cutoff is used.

ELISA is still preferable in equipped central laboratories [15,18–20]. FIA needs power (or a charged analyzer) and cartridges: it is **near-POC**, not zero-infrastructure. National tiers should be: visual ICT or FIA at the periphery; ELISA and RT-PCR at the reference laboratory; composite algorithms for severe or discordant cases [2,4,30].

Proposed algorithm (refine after NLR is known).

- (1) Days 0–3, reader available: FIA NS1 first-line; positive → dengue pathway; confirm with ELISA/RT-PCR as capacity allows.
- (2) Days 4–7 or suspected secondary infection: FIA plus IgM ($\pm$ IgG); dual-negative → RT-PCR or paired serology [9,20,29].
- (3) Severe dengue, ICU, or surveillance confirmation: RT-PCR and/or NS1 ELISA [20,30].
- (4) Negative FIA, high clinical probability: never a stop test.

4.3 Strengths

The question is technology-specific: Mata and Haider pooled visual ICTs [28,29]. Reporting follows PRISMA-DTA/S and dual independent screening/extraction [31–34]. QUADAS-2 was adapted for fever timing, IFU, indeterminates, incorporation bias, and diagnostic window [36,39]. Pooling used Reitsma/HSROC rather than univariate Se/Sp or DOR-only models [42–44], with Deeks’ test [45] and pre-specified NS1-biology covariates [9,12,14]. The paired contrast avoids case-mix confounding. Analysis code is archived.

4.4 Limitations

RT-PCR, isolation, composite algorithms, and NS1 ELISA are not interchangeable. ELISA-only reference can inflate agreement (incorporation bias) [39]; RT-PCR after viremia can lower apparent FIA sensitivity [9,20]. We did not use latent-class models as the primary analysis [20]. Fever day is often reported as a mixed mean, blurring days 0–3 vs 4–7. Serotype-specific 2×2 tables are scarce; DENV-4 and DENV-2 have been associated with lower NS1 ELISA sensitivity [18,19]. Flavivirus cross-reactivity (Zika, occasionally YFV/JEV) can inflate false positives if controls are blood donors rather than other acute flavivirus infections [17,30]. IgM/IgG ratios imperfectly classify immune status. Lot, reader model, and sponsorship were incompletely reported. Expected k is smaller than ICT reviews [28,29], so prediction regions may be wide. Continuity correction is conventional but arbitrary. Language bias may persist despite no language filter. This review did not meta-analyse cost, COI–severity, combined NS1+IgM cartridges (unless NS1 was separable), donor screening, or pre-vaccination IgG [30].

4.5 Future research and policy

Needed: prospective STARD-compliant paired FIA/ELISA studies with independent RT-PCR and day-of-illness to the day [36,51]; powered secondary-infection strata; regional flavivirus challenge panels [17,30]; lot and bor-

derline-COI reporting; impact studies (time-to-decision, admission, antibiotics, severe dengue); four-way FIA vs ICT vs ELISA vs RT-PCR; vaccine-era strata in which NS1 may outlast IgM/IgG for acute diagnosis [30].

Policy: do not tender “rapid dengue tests” as one class. Place FIA in near-POC triage; do not use a negative FIA as standalone rule-out. Keep ELISA/RT-PCR for confirmation, serotyping, and EQA of peripheral FIA [2,20]. State in case definitions whether FIA NS1 is confirmatory or screening.

4.6 Conclusions

Early NS1 detection in the first week of fever is central to dengue care. ELISA’s accuracy is offset by delay; visual ICT’s speed is offset by modest, operator-dependent sensitivity [17,18,22,28,29]. FIA was built for that gap.

In this PRISMA-DTA review, FIA pooled sensitivity was [xx.x%] (95% CI [ll]–[uu]) and specificity [xx.x%] ([ll]–[uu]), with DOR [xxx] and AUC [0.xx]. Versus ELISA on paired specimens, FIA was [fill from Table 7]. Performance [was / was not] worse in secondary infection and [was / was not] better in the first three days of illness [9,12,14].

Actionable points. (1) Use FIA as a rapid, objective NS1 **rule-in** test in triage, emergency, and outbreak settings when a reader can be maintained—especially days 0–3. (2) Do not use a negative FIA to exclude dengue in secondary infection, late acute presentation, or severe disease; reflex to IgM/IgG, ELISA, and/or RT-PCR. (3) Do not equate FIA with visual ICT or ELISA in guidelines or tenders. (4) Keep ELISA/RT-PCR as the laboratory reference. (5) Commission paired, day-stratified, serotype-aware studies with independent molecular reference before making FIA the sole confirmatory test.

Until that evidence accumulates, FIA is a clinically useful **near-point-of-care NS1 assay** that can shorten time-to-diagnosis toward ELISA-like objectivity without fully replacing laboratory confirmation.

Acknowledgements

[To be completed.] We thank [librarian name] for PRESS review of the search strategy, if applicable.

Funding

[Declare grants, or: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.]

Conflicts of interest

[Declare, or: The authors declare no competing interests.]

Author contributions (CRediT)

Conceptualization: []; Methodology: []; Software: []; Formal analysis: []; Investigation: []; Data curation: []; Writing — original draft: []; Writing — review and editing: []; Visualization: []; Supervision: []; Project administration: [].

Data availability

Extracted 2×2 tables, QUADAS-2 judgements, and R scripts will be deposited at [URL] upon acceptance. Search strategies and extraction templates accompany this manuscript as supplementary files.

Ethics

Not required. This study used published aggregate data only.

Figure legends

- Figure 1.** PRISMA 2020 flow diagram of identification, screening, eligibility, and inclusion.
- Figure 2.** Geographic distribution of included studies (bubble size proportional to sample size).
- Figure 3.** QUADAS-2 traffic-light plot (study × domain).
- Figure 4.** QUADAS-2 weighted bar chart of Low, High, and Unclear judgements.
- Figure 5.** Coupled forest plots of FIA sensitivity and specificity.
- Figure 6.** HSROC curve for FIA (summary point, 95% confidence region, 95% prediction region).
- Figure 7.** Comparative HSROC: FIA versus ELISA on paired specimens.
- Figure 8.** Deeks’ funnel plot of $1/\sqrt{\text{ESS}}$ versus $\ln(\text{DOR})$.

Supplementary files

File	Content
S1	PRISMA-DTA checklist
S2	Full database-specific search strategies
S3	Data extraction template
S4	Tailored QUADAS-2 protocol and decision rules
S5	R analysis script (dta_meta_analysis.R)
S6	Study-level QUADAS-2 judgements; GRADE (if applied); ELISA and agreement 2×2 tables

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Pooled estimates from this review were not fabricated. Quantitative Results cells are to be filled after search, dual extraction, QUADAS-2, and bivariate modelling.